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REPORT WRITING

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CYBERSECURITY THREATS AND HOW TO SOLVE IT

1.0 INTRODUCTION

The growing reliance on the Internet for daily tasks from communication to shopping has led to a rapidly expanding digital ecosystem. As Burns (2018) noted, the surge in e-commerce reflects users' willingness to trade privacy for convenience. This digital transformation, however, has introduced a range of cybersecurity threats, including phishing, data breaches, insider threats, and vulnerabilities in technologies like the Internet of Things (IoT). Cybercriminals now exploit weak points in data transmission, targeting both secure and unsecured networks, with phishing attacks becoming increasingly personalized (Sathish, 2018). Insider threats also pose serious risks, especially when sensitive data or intellectual property is involved. Fortinet (2023) warns that unsecured IoT devices are particularly vulnerable, offering gateways to large-scale attacks. These challenges highlight a growing concern: both individuals and organizations face increasing cyber threats due to low awareness and outdated security practices. As online activity grows, so does the risk of financial, reputational, and operational damage. This report will be focusing on the cause of cyberthreats, what the effects are and how to solve it.

2.0 PROBLEM'S BACKGROUND

In today's world, more people and companies are using digital devices and online services. This makes it easier to work, communicate and store information. However, it also opens the door to more cyber threats. Hackers and cybercriminals now have more chances to attack systems, steal private data and cause damage. Cybersecurity has become a major concern for everyone, not just big companies. One of the main reasons for this is the growing use of Internet of Things (IoT) devices like smart TVs, security cameras and even smart home appliances. Many of these devices have weak protection, which makes them easy targets for hackers. At the same time, people are still falling for phishing emails or clicking on suspicious links. Someone, even people inside an organization might leak important data by mistake or on purpose. Even after a service or platform gets hacked, some users continue to use it without changing passwords or updating their information. This shows that people are either unaware of the risk or feel helpless to do anything about it. All these issues show that we need to improve both our technology and our awareness when it comes to online safety.

2.1 CAUSED OF PROBLEM

Cybersecurity problems are mostly caused by weak security practices and a lack of awareness among users. Many IoT devices still use default settings or weak passwords This report investigates the causes and consequences of cyber threats and explores effective strategies to protect digital assets in today's high-risk online environment.

ds, making them easy targets for hackers. Users also often fall for phishing attacks because they do not fully understand how to spot fake emails or links. Some threats come from within organizations, such as employees who misuse their access or accidentally leak data. In addition, many systems are outdated and not equipped to handle modern cyber threats. Even after a breach, people tend to continue using the same services without changing passwords or updating their devices, which increases the risk of future attacks.

2.2 EFFECT OF THE PROBLEM

One effect of cyber threats is the significant damage they cause to organizational networks and data systems. Vulnerabilities in IoT devices, for instance, allow attackers to breach a single device and move laterally across networks to access sensitive corporate data or spread malware (Fortinet, 2023). This disrupts operations and exposes businesses to financial and reputational losses. In more severe cases, cybercriminals use infected devices to form botnets and launch Distributed Denial-of-Service (DDoS) attacks, such as the 2016 Mirai incident that took down major websites. Another critical effect is financial loss and widespread disruption caused by phishing campaigns, where victims are tricked into revealing personal data or downloading malware like trojans and worms (Sathish, 2018). These attacks are difficult to trace and extremely costly, with annual U.S. losses estimated between $100 million and $3 billion. Financial and banking services are especially targeted, making up nearly 93% of reported phishing attacks. Overall, the growing impact of cyber threats highlights the urgent need for stronger security awareness and protection.

3.0 POSSIBLE SOLUTIONS

To reduce the number of cybersecurity problems, both users and organizations need to take action. We can not rely only on old security methods. We need to use updated tools, improve user behavior and make sure people are more aware of how to stay safe online.

3.1 SOLUTION 1

One of the first things to fix is the security of smart devices and networks. People should always change default passwords and use stronger ones. Devices should also be updated regularly with the latest software to fix security bugs. It is also helpful to separate smart devices from the main network. For example, if someone hacks a smart TV, they should not be able to access your computer files. Firewalls today also come with smarter features that can detect and stop threats faster using AI and machine learning. At a bigger level, like in companies or schools, IT teams should monitor the network closely and limit who gets access to certain systems. These steps may sound small, but they can stop a lot of common attacks before they cause serious damage.

3.2 SOLUTION 2

To protect users from physical threats, there is a solution also for physical security known as the computing devices control. This solution involves users themselves, training users to protect their devices from various ways, thus it is as important as traditional technical solutions. Usually devices are facing risk from threats like physical tampering, side-channel attacks, fault injection and unauthorized access to the debug interfaces. To meet these challenges, preventive measures including tamper-resistance, shielding information and disabling access ports are implemented . Involving sensors that able to detect light or motion and trigger responses like data losing or system shutdown suddenly will help to protect users from malicious tampering (Weingart, 2000). Besides that, users can have multiple internal layouts at the same time to confuse the attackers, disabling unused ports and hiding debug interfaces to reduce attacks surface.

4.0 CONCLUSION

Cybersecurity has become more important as the growing popularity of internet users in the world. As this report shows that threats like phishing, hacking and data leaks have become more common due to the increasing users of computing devices and Internet of Things (IoT). These threats can cause loss of money, giving away personal information, and even physical damage to an individual or a company. These threats can happen usually due to people nowadays not aware of the risks and do not solve these problems seriously to protect themselves. It is important to ensure our internet environment stays safe by using strong passwords, keeping software up-to-date and learn how to prevent online scamming. Companies also need to train their employees to secure their devices properly while using them. By taking these simple steps, we can make the internet safer in the future digital world.

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